

MBB assortativity — empirical sorting, pools, and PPM (SCOUT)

Prepared: September 25, 2026
For: Charles / VECTOR
Lane: SCOUT — saved artifacts, repo paths, execution gaps.
Status: Read-only. No runs in this note.
Use with (same folder, separate memos): VECTOR (interpretation, math, bounded audit); COMPASS (framing, hypotheses, campaign).
Hygiene / PPM construction: SCOUT data-hygiene response §3.6.

Evidence tags

Tag	Meaning
[Direct repo]	Code, config, or file in repository today
[Saved result]	Named artifact; not re-run here
[Meeting record]	Charles–Alex conversation Sep 25, 2026 (summary only)
[Gap]	Not in repo or blocked

1. Charles’s question (pointer)

[Meeting record] Pool composition, PPM, five-on-court scarcity, Squid vs Jackals, and “why measured assortativity ≈ 0 ?” — recorded in VECTOR and COMPASS memos. SCOUT below is **artifact and pipeline facts** only.

2. Estimand map (repo objects)

[Direct repo] BINDING_Selection_is_its_own_step.md ; GRANDCHILD_D_and_H_sort_interpretation.md ; pd21_rho_hsort_calibrate.py ; panel_rebuild.py ; panel_build.py .

Object	Code / artifact
H_{sort}	realized_sorting_index_H_sort / PD21 empirical_h_sort
ρ^* bracket	PD21_rho_hsort_calibrate_*_fit_bracket.json
poolq_loo	recompute_teammate_loo_pool_quality after assign_perf_from_metric("ppm")
Assort sim (paused)	ASSORT_20260925_initial_execution_stop.md

3. Saved H_{sort} and panel provenance

[Saved result] PD21 brackets, PPM $\rightarrow$ within-season z, min20 drop:

Artifact	Generated	2015 empirical H_{sort}	Provenance note
PD21_rho_hsort_calibrate_2015_fit_bracket.json	2026-08-14	≈ 0.114	Before mg10 lock (Aug 17) — not current QC panel
PD21_rho_hsort_calibrate_2013_2021_fit_bracket.json	2026-08-19	≈ 0.061 (2015 row)	Post-QC era trace; $\rho = 0$ sim mean ≈ 0.081 same row
Same 2013–2021 file	—	$\approx 0.056\text{--}0.071$ other seasons	Per-season trace in JSON

[Direct repo] mg10: BOX_QC_panel_build_policy.md §3b; config.min_team_season_games = 10 .

[Saved result] Grandchild $\rho \rightarrow D$ (2015, $C = 15$): GRANDCHILD_D_and_H_sort_interpretation.md .

How to read magnitudes (z-score vs H_{sort} , random-allocation null, LOO vs partition): VECTOR memo — not duplicated here.

4. Panel counts and PPM facts (Sep 25 rebuild)

[Saved result] build_from_box , mg10 + dash + min20, all seasons in box:

Quantity	Value
Player-season rows	88,399 (2005–2025)
Team-seasons	7,325
Mean eligible roster size	≈ 12.1
2015 team-seasons	351
2015 mean roster size	≈ 12.2

Zero PPM after min20 (PPM = 0 ⇔ 0 points, ≥20 season minutes):

Scope	Mean per team-season	Share of teams with ≥1
All seasons	≈ 0.05	≈ 4.6%
2015	≈ 0.08	≈ 7.7%

[Direct repo] PPM not in game CSV; computed at aggregation — [hygiene §3.6](#).

5. Hero / QC artifacts (tail, not H_{sort})

[Saved result] Post-QC Pass A pair and CPR tables: [hygiene response](#) §§2, 7; `reigning_hero/README.md` (different spec: last-PS, EW16, 09–21).

[Saved result] Sep 2026 perf-metric LOO batch — PPM retained for hero shape gate: `sports_sandbox/_DISPOSABLE_perf_metric_rho_eda/LOO_SHAPE_REPORT.md` (path per campaign docs).

6. Checklist → evidence location (SCOUT column only)

Charles question	Repo fact
Metric	PPM z; PD21 JSON §3; hygiene §3.6
Min20	<code>panel_rebuild</code> after aggregation; production lock in hygiene §3
Mg10	<code>panel_rebuild._apply_box_qc</code> ; Pass A sensitivity README
D-I only	[Gap] no season-specific membership crosswalk in panel builder
Draft-ever teams	<code>restrict_teams_by_draftees</code> in config — hero path; not assort main spec
Assort experiment	Stopped — execution stop

7. Gaps and blocked work (SCOUT)

- 1. 2015 assort population — canonical-team + partial points/minutes rows unresolved ([execution stop](#)).
- 2. Per-game rotation audit — not run; proposed in VECTOR memo.
- 3. Permutation H_{sort} null on frozen population — not in repo.
- 4. Transfer rows at `(athlete_id, season, team_id)` — [Gap] no saved 2015 audit vs last-team-only rule.
- 5. Assort 100× repetitions — not started.

Next steps / hypotheses / audit design: VECTOR and COMPASS memos.

Thread log

- 2026-09-25 — Initial SCOUT artifact note.
- 2026-09-25 (pm) — Trimmed to SCOUT-only facts; interpretation lives in VECTOR + COMPASS.

SCOUT — evidence and gaps. Read all three memos in `docs/scientific_questions/`.